

Cedric Allier

Plexus

A vesicle that grows petals where the activator is high



entities

cell 2,000 of 65,004 declared
vertex 130,004
↔ half_edge 520,032 vertex → vertex

activities

 seed
build the cell mesh
initial morphogen
 $u_j = 1$ everywhere, $a_j = 0$ but on the seeded patch

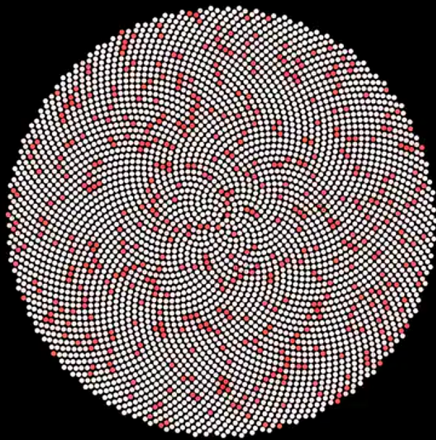
 aggregate
cell geometry from the mesh
$$\mathbf{cen}_f = \frac{1}{n_f} \sum_{e \in f} \mathbf{x}_{\text{src}}(e), \quad A_f = \frac{1}{2} \left\| \sum_{e \in f} \mathbf{x}_{\text{src}}(e) \times \mathbf{x}_{\text{tgt}}(e) \right\|$$

 rewire
who neighbours whom
T1 neighbour exchange

 lateral
morphogen diffusion between neighbouring cells
autocatalytic reaction
chemistry driven by curvature
growth where the morphogen is high
vertex-model shape energy
purse-string line tension

 divide
cell division

Gray-Scott chemistry on a disc of 4,000 cells



entities

cell 4,000

activities



seed

initial positions

$\mathbf{x}_i = \text{placement}(\text{mode}, r, i)$

initial morphogen

$u_j = 1$ everywhere, $a_j = 0$ but on the seeded patch



rewire

neighbours within a radius

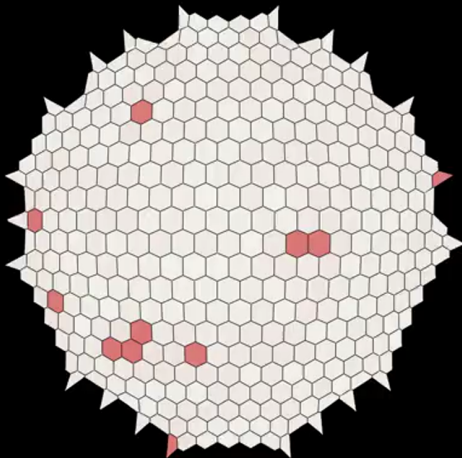


lateral

morphogen diffusion between neighbouring cells
autocatalytic reaction

A tissue that remodels itself: the vertex model

sheet_morphogen_die mesh frame 1/60 374 cells



entities

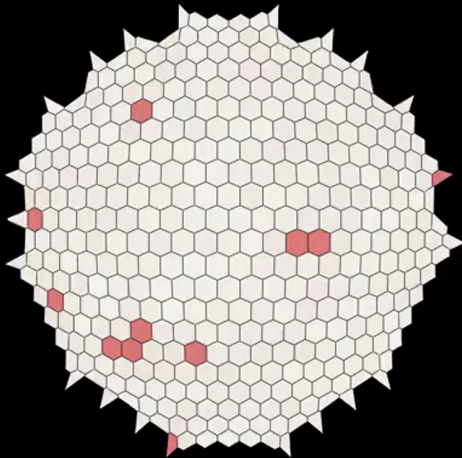
cell 400 of 2,000 declared
vertex 4,000
↔ half_edge 16,000 vertex → vertex

activities

- seed
 - build the cell mesh
 - initial morphogen
 - $u_j = 1$ everywhere, $a_j = 0$ but on the seeded patch
- aggregate
 - cell geometry from the mesh
 - $\text{cen}_f = \frac{1}{n_f} \sum_{e \in f} \mathbf{x}_{\text{source}}(e)$, $A_f = \frac{1}{2} \left\| \sum_{e \in f} \mathbf{x}_{\text{source}}(e) \times \mathbf{x}_{\text{target}}(e) \right\|$
- rewire
 - who neighbours whom
 - T1 neighbour exchange
- lateral
 - morphogen diffusion between neighbouring cells
 - autocatalytic reaction
 - growth where the morphogen is high
 - vertex-model shape energy
- die
 - cell death

A tissue that remodels itself: the vertex model

sheet_morphogen mesh frame 1/60 374 cells



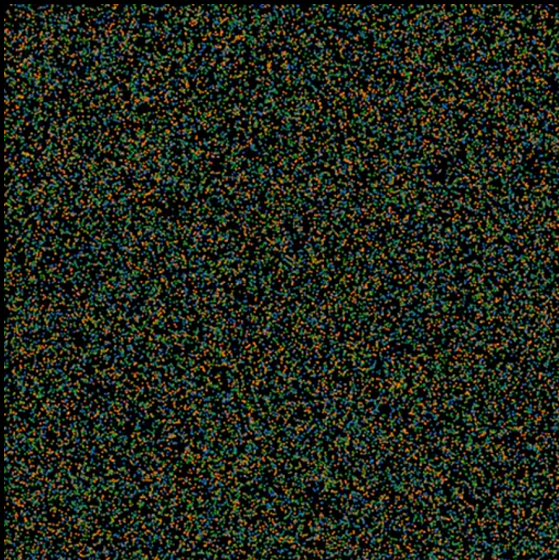
entities

cell 400 of 2,000 declared
vertex 4,000
↔ half_edge 16,000 vertex → vertex

activities

-  seed
build the cell mesh
initial morphogen
 $u_j = 1$ everywhere, $a_j = 0$ but on the seeded patch
-  aggregate
cell geometry from the mesh
$$\mathbf{cen}_f = \frac{1}{n_f} \sum_{e \in f} \mathbf{x}_{\text{src}}(e), \quad A_f = \frac{1}{2} \left\| \sum_{e \in f} \mathbf{x}_{\text{src}}(e) \times \mathbf{x}_{\text{tgt}}(e) \right\|$$
-  rewire
who neighbours whom
T1 neighbour exchange
-  lateral
morphogen diffusion between neighbouring cells
autocatalytic reaction
growth where the morphogen is high
vertex-model shape energy

Three particle types under one pairwise force law



entities

particle 30,000 t0, t1, t2

activities

 rewire

neighbours within a radius

$$E = \{ (i, j) : r_{\min} \leq \| \mathbf{x}_i - \mathbf{x}_j \| < r \}$$

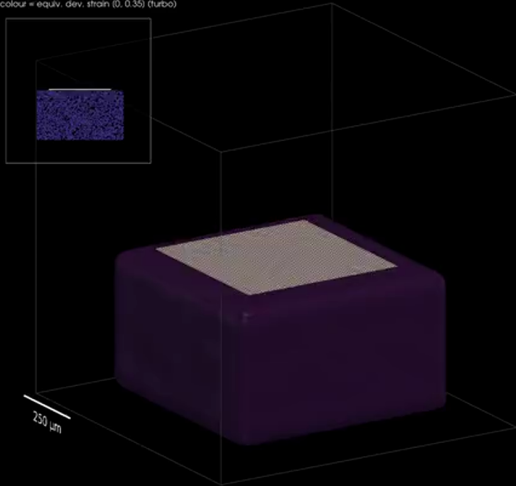
 lateral

attraction and repulsion

$$f(r) = p_1 e^{-r^2 p_2 / 2 \sigma^2} - p_3 e^{-r^2 p_4 / 2 \sigma^2}, \quad \dot{\mathbf{x}}_i = \sum_{j \in N(i)} f(r_{ij}) (\mathbf{x}_j - \mathbf{x}_i)$$

An elastic sheet pressed into a gel of 500,000 material points

mesh mpm_sheet_elastic box 1 mm cube
500,000 nodes grid 64^3, dx 15.6 μm
frame 21/300 275 ms/frame compute
t = 0.0064 s 4x slow
colour = equiv. dev. strain (0, 0.35) (turbo)



entities

vertex	4,096	
↔ half_edge	18,432	vertex → vertex
plate_face	2,304	
mpm_particle	500,000	gel
mpm_grid		field, grid 64

activities



seed

lay out the plate



structural

drive the surface at a set rate

$$x_a(t) \leftarrow x_a(0) + b_y \text{ clamp}\left(\frac{t - t_{\text{hold}}}{t_{\text{over}}}, 0, 1\right), \quad v_a = \frac{b_y}{t_{\text{over}}}$$



lateral

contact with a live surface

$$\mathbf{a}_n = k \, d_f \, \mathbf{n}, \quad \mathbf{v}_f = (\mathbf{v}_f - \mathbf{v}_{\text{face}}) - ((\mathbf{v}_f - \mathbf{v}_{\text{face}}) \cdot \mathbf{n}) \mathbf{n}, \quad \mathbf{a}_f = -\mu \|\mathbf{a}_n\| \frac{\mathbf{v}_f}{\|\mathbf{v}_f\| + \epsilon_v}$$

material update



exchange

particle to grid

grid to particle

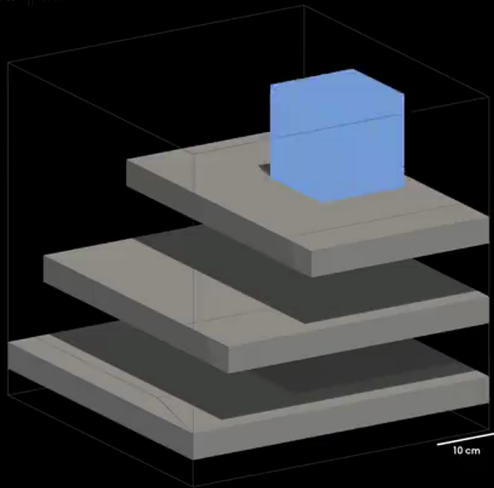


field

grid solve

A waterfall of 5,000,000 material points

sl waterfall_5m_box 50 cm cube
5,000,000 particles grid 192³ = 7.1M cells, dx 2.6 mm
frame 8,2920 3388 ms/frame compute
t = 0.006667 s 4x slow



entities

cell 5,000,000 water
mpm_grid field, grid 192

activities



lateral

gravity

$\mathbf{a}_{\text{ext}} = \mathbf{g}$ (default $\mathbf{g} = (0, -g_y)$)

material update

$\mathbf{F} \leftarrow (I + \Delta t \mathbf{C}) \mathbf{F}$ (liquid: $\mathbf{F} \leftarrow J^{1/d} \mathbf{I}$, $J = \det \mathbf{F}$)

shear viscosity



exchange

particle to grid

grid to particle



field

grid solve

60,000 particles riding the Halvorsen attractor

entities

cloud 60,000 c

activities



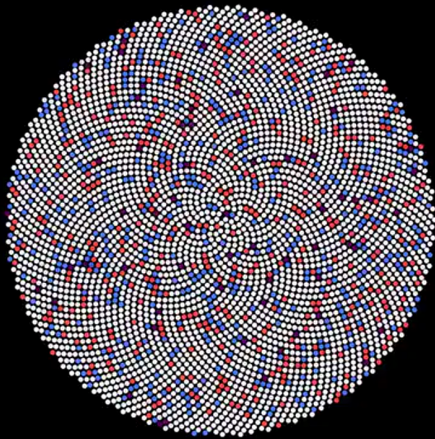
lateral

flow along a strange attractor

$\dot{\mathbf{x}} = f_{\text{system}}(\mathbf{x})$, e.g. Lorenz: $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$



Two coupled chemistries on a disc of 4,000 cells



entities

cell 4,000

activities

 seed

initial positions

$\mathbf{x}_i = \text{placement}(\text{mode}, r, i)$

initial morphogen $\times 2$

$u_j = 1$ everywhere, $a_j = 0$ but on the seeded patch

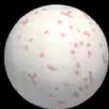
 rewire

neighbours within a radius

 lateral

morphogen diffusion between neighbouring cells $\times 3$
autocatalytic reaction

A vesicle growing arms, with the purse-string tension restored



entities

cell 2,000 of 65,004 declared
vertex 130,004
↔ half_edge 520,032 vertex → vertex

activities



seed

build the cell mesh

initial morphogen

$u_j = 1$ everywhere, $a_j = 0$ but on the seeded patch



aggregate

cell geometry from the mesh

$$\mathbf{cen}_f = \frac{1}{n_f} \sum_{e \in f} \mathbf{x}_{\text{seed}}(e), \quad A_f = \frac{1}{2} \left\| \sum_{e \in f} \mathbf{x}_{\text{seed}}(e) \times \mathbf{x}_{\text{right}}(e) \right\|$$



rewire

who neighbours whom

T1 neighbour exchange



lateral

morphogen diffusion between neighbouring cells

autocatalytic reaction

chemistry driven by curvature

growth where the morphogen is high

vertex-model shape energy

purse-string line tension

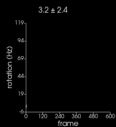
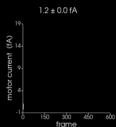
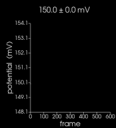
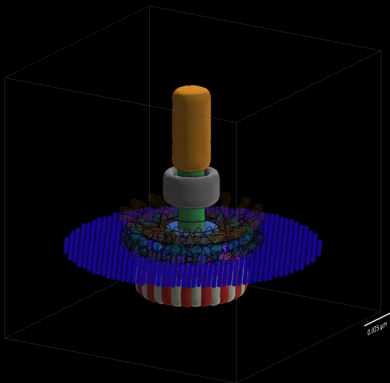


divide

cell division

The bacterial flagellar motor, as 415,000 material points

bfm_e01t_cj_membrane_avg_box 0.14 μm cube
51,632 nodes grid 128/3, dx 0.001 μm
frame 3/60 1547 ms/frame compute



entities

c_posts_a	73,759	
c_posts_b	73,759	
c_flg	51,632	
ms_ring	35,137	
rod	25,566	
lp_ring	18,555	
hook	62,234	
cell	1	
↪ stator_unit	17	17 per cell
↪ membrane	1,656	1,656 per cell
Fill	6,664	
PtiA	13,141	
PtiB	11,611	
PtiCD	30,107	
Lipo	5,406	
MotB	6,137	
mpm_grid		field, grid 128

activities

- seed
 - place a measured point cloud × 8
 - $\mathbf{x}_i = \mathbf{x}_{orig} + \text{scale} \cdot \mathbf{q}_i$
- lateral
 - membrane potential

$$C \frac{d\psi}{dt} = j_{\text{pump}}(\psi, \text{ssr} = \text{"Membranepotentialhomrespiration"}) - j_{\text{leak}} - g\psi, \quad j_{\text{pump}} = j_{\text{max}} \max\left(0, 1 - \frac{\psi}{\psi_{\text{max}}}\right)$$
 - stator step
 - mpm anchor
 - material update × 13
- broadcast
 - broadcast × 3
- exchange
 - stator-to-rotor linkage
 - ion current into the membrane
 - particle to grid × 13
 - grid to particle × 13
- aggregate
 - sum the children onto their parent
- field
 - grid solve